

ANALYTICAL PROBLEM-SOLVING ACTIVITY

Topic 1. Metrics for Performance Evaluation

1. Accuracy

A classification model tested **100 samples**. It correctly classified **85 samples**.

Questions:

1. How many samples were classified incorrectly?
2. Calculate the accuracy of the model.
3. Calculate the error rate.
4. Is the model performing well based on accuracy?
5. What happens to accuracy if 5 more samples are classified correctly?

ANSWER:

1. Accuracy

Given: Total samples = 100, Correctly classified = 85

1. Incorrectly classified samples

$$100 - 85 = 15$$

1. Accuracy

$$\frac{85}{100} \times 100 = 85\%$$

Answer: 85%

1. Error rate

$$\frac{15}{100} \times 100 = 15\%$$

1. Is the model performing well?

Yes. An accuracy of 85% is generally considered good.

1. If 5 more samples are classified correctly

Correct predictions = 90

$$\frac{90}{100} \times 100 = 90\%$$

Answer: 90% accuracy

2. Confusion Matrix

A model gives the following results:

- True Positive (TP) = 40
- True Negative (TN) = 35

- False Positive (FP) = 10
- False Negative (FN) = 15

Questions:

1. What is the total number of predictions?
2. Calculate the accuracy.
3. Calculate the error rate.
4. Which value represents incorrectly predicted positive cases?
5. Which value represents incorrectly predicted negative cases?

ANSWER:

Given: TP = 40, TN = 35, FP = 10, FN = 15

1. Total predictions

$$40 + 35 + 10 + 15 = 100$$

Answer: 100

1. Accuracy

$$\frac{40+35}{100} \times 100 = 75\%$$

Answer: 75%

1. Error rate

$$\frac{10+15}{100} \times 100 = 25\%$$

Answer: 25%

1. Incorrectly predicted positive cases

Answer: False Positive (FP = 10)

1. Incorrectly predicted negative cases

Answer: False Negative (FN = 15)

3. Precision and Recall

A disease prediction system produces:

- TP = 80
- FP = 20
- FN = 10
- TN = 90

Questions:

1. Calculate Precision.
2. Calculate Recall.
3. Which metric tells us how many predicted positive cases were actually positive?
4. Which metric tells us how many actual positive cases were correctly identified?
5. Which metric is more important when missing a positive case is dangerous?

ANSWER:

Given: TP = 80, FP = 20, FN = 10, TN = 90

1. Precision

$$\frac{80}{80+20} = \frac{80}{100} = 0.8 = 80\%$$

Answer: 80%

1. Recall

$$\frac{80}{80+10} = \frac{80}{90} = 0.889 = 88.9\%$$

Answer: 88.9%

1. Predicted positive cases that were actually positive

Answer: Precision

1. Actual positive cases correctly identified

Answer: Recall

1. More important when missing a positive case is dangerous

Answer: Recall

4. Compare Two Classifiers

Two classifiers are tested on the same dataset.

Metric	Classifier A	Classifier B
Accuracy	90%	85%
Precision	80%	95%

Recall	70%	90%
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Questions:

1. Which classifier has higher accuracy?
2. Which classifier has higher precision?
3. Which classifier has higher recall?
4. Which classifier would you choose if detecting as many positive cases as possible is important?
5. Give one reason why accuracy alone may not be enough.

ANSWER:

1. Higher accuracy: Classifier A
2. Higher precision: Classifier B
3. Higher recall: Classifier B
4. Choose for detecting as many positive cases as possible: Classifier B
5. Why accuracy alone is not enough?

Because a model can have high accuracy but still miss many positive cases.

5. Analyze Model Performance

A classification model produces the following results:

- TP = 70
- TN = 80
- FP = 20
- FN = 30

Questions:

1. Calculate accuracy.
2. Calculate precision.
3. Calculate recall.
4. Calculate the error rate.
5. Based on these values, identify one strength and one weakness of the model.

ANSWER:

Given: TP = 70, TN = 80, FP = 20, FN = 30

1. Accuracy

$$\frac{70+80}{200} \times 100 = 75\%$$

Answer: 75%

1. Precision

$$\frac{70}{70+20} = \frac{70}{90} = 77.8\%$$

Answer: 77.8%

1. Recall

$$\frac{70}{70+30} = \frac{70}{100} = 70\%$$

Answer: 70%

1. Error rate

$$\frac{20+30}{200} \times 100 = 25\%$$

Answer: 25%

1. Strength and weakness

- Strength: Good precision (77.8%)
- Weakness: Recall is only 70%, so some positive cases are missed

Topic 2. Prediction Algorithms

1. Simple Linear Prediction

A company's sales are predicted based on advertising expenditure.

Advertising Cost (₹)	Sale s
10	20
20	40
30	60
40	80

Assume the relationship is linear.

Questions:

1. Identify the relationship between advertising cost and sales.
2. Predict sales when advertising cost is ₹50.
3. What happens to sales when advertising cost increases by ₹10?
4. Is this an example of classification or prediction?
5. Why is a linear prediction algorithm suitable here?

ANSWER:

Relationship: Sales = 2 × Advertising Cost

1. Relationship: Direct linear relationship.
2. Advertising = ₹50

Sales = 100

1. Increase of ₹10

Sales increase by 20 units.

1. Classification or prediction?

Prediction

1. Why linear prediction is suitable?

Because sales increase proportionally with advertising cost.

2. Predict Temperature

A simple prediction model is:

Questions:

1. Predict the temperature on Day 5.
2. Predict the temperature on Day 10.
3. What is the predicted temperature on Day 15?
4. How much does the temperature increase for every additional day?
5. Is this a classification or prediction problem?

ANSWER:

Assume the model is:

$$T = 20 + 2(\text{Day})$$

1. Day 5

$$20 + 2(5) = 30^{\circ}\text{C}$$

1. Day 10

$$20 + 2(10) = 40^{\circ}C$$

1. Day 15

$$20 + 2(15) = 50^{\circ}C$$

1. Increase per day

2°C

1. Classification or prediction?

Prediction

3. House Price Prediction

A prediction algorithm estimates house prices based on area.

Area (sq.ft)	Price (₹ lakhs)
1000	40
1500	60
2000	80

Assume a linear relationship.

Questions:

1. Identify the relationship between area and price.
2. Predict the price of a 2500 sq.ft house.
3. What happens to the price when area increases by 500 sq.ft?
4. Which type of algorithm can be used for this problem?
5. Is this classification or numerical prediction?

ANSWER:

1. Relationship

Price increases linearly with area.

1. 2500 sq.ft

Price = ₹100 lakhs

1. Increase of 500 sq.ft

Price increases by ₹20 lakhs

1. Algorithm

Linear Regression

1. Classification or numerical prediction?

Numerical prediction

4. Compare Actual and Predicted Values

A prediction algorithm gives the following results:

Actual Value	Predicted Value
50	48
60	65
70	68
80	85
90	88

Questions:

1. Which prediction has the smallest error?
2. Which prediction has the largest error?
3. Calculate the error for each prediction.
4. Is the model generally predicting close to the actual values?
5. Why is the difference between actual and predicted values important?

ANSWER:

1. Smallest error

2 (first, third, and fifth predictions)

1. Largest error

5 (second and fourth predictions)

1. Errors

- $50 \rightarrow 48 = 2$
- $60 \rightarrow 65 = 5$
- $70 \rightarrow 68 = 2$
- $80 \rightarrow 85 = 5$
- $90 \rightarrow 88 = 2$

1. Is the model predicting close to actual values?

Yes, the errors are small.

1. Why is the difference important?

It shows the prediction accuracy of the model.

5. Choosing a Prediction Algorithm

Consider the following applications:

- A. Predicting tomorrow's temperature.
- B. Predicting the price of a house.
- C. Predicting the number of products sold next month.
- D. Predicting whether an email is spam or not.

Questions:

1. Which applications are **prediction problems**?
2. Which application is a **classification problem**?
3. Which algorithm could be used for predicting house prices?
4. Why is regression suitable for numerical prediction?
5. Give one difference between **classification** and **prediction**.

ANSWER:

Applications:

- A. Tomorrow's temperature
- B. House price
- C. Products sold next month
- D. Email spam

1. Prediction problems

A, B, and C

1. Classification problem

D (Spam detection)

1. Algorithm for house prices

Linear Regression

1. Why regression is suitable?

Because it predicts continuous numerical values.

1. Difference between classification and prediction
 - Classification: Predicts categories (Spam / Not Spam).
 - Prediction (Regression): Predicts numerical values (Price, Temperature, Sales).

